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INFORMATION SYSTEMS RESEARCH
Vol. 30, No. 2, June 2019, pp. 507–522
ISSN 1047-7047 (print), ISSN 1526-5536 (online)

Effects of Sponsorship Disclosure on Perceived Integrity of Biased Recommendation Agents: Psychological Contract Violation and Knowledge-Based Trust Perspectives

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Received: April 18, 2016
Revised: May 1, 2017; February 26, 2018; June 1, 2018
Accepted: July 8, 2018
Published Online in Articles in Advance: May 24, 2019

https://doi.org/10.1287/isre.2018.0811

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Abstract. The perceived integrity of online product recommendation agents (RAs) becomes a critical trust concern when RAs apply sponsorship practices and recommend products *biased* toward sponsored products. Sponsorship disclosure is enforced by the U.S. Federal Trade Commission, but many technologies fail to comply, probably because of their concerns about users' trust in the biased technologies. By applying the theoretical perspectives of psychological contract violation and knowledge-based trust, we intend to reveal when sponsorship disclosure is most effective in enhancing users' perceived RA integrity. A laboratory experiment revealed two major findings. First, sponsorship disclosure leads to users' perceived RA integrity via reduced perceived psychological contract violations of a biased RA, but only for users with high prior knowledge about the prevalence of sponsorships used by RAs in general. For users with limited prior knowledge, the disclosure fails to reduce these perceived violations. Second, regardless of the level of such prior knowledge of users, sponsorship disclosure enhances users' perceived transparency of a biased RA, which, in turn, leads to perceived RA integrity. Theoretical and practical implications are discussed.

History: Paul A. Pavlou, Senior Editor; Animesh Animesh, Associate Editor.
Funding: The work described in the paper was partially supported by grants from the Research Grants Council of Hong Kong S.A.R. [Projects CityU 11505917 and CityU 148209], the National Natural Science Foundation of China (NSFC) [Project 71471156], and the Shenzhen Research Institute of the City University of Hong Kong.

Keywords: recommendation agent • neutrality • bias • sponsorship • integrity • trust • psychological contract violation • transparency

1. Introduction

Product recommendation agents (RAs), especially content-filtering-based RAs, are widely employed by online merchants to facilitate consumers' decision making for complex products, such as electronics, automobiles, and other configurable industrial products (e.g., Li and Karahanna 2015, Panniello et al. 2016, Wang and Benbasat 2016, Adomavicius et al. 2018). Content-filtering-based RAs elicit and identify users' preferences to generate recommendations. Extant behavioral research on these RAs primarily focuses on *neutral* RAs that recommend products based solely on users' expressed preferences (i.e., organic recommendations) and sort these products according to their fit to user preferences (e.g., Xiao and Benbasat 2014). In practice, however, RAs usually produce *biased* recommendations in addition to organic ones. Biases occur when an RA "systematically favors a small set of items over other items in the database irrespective of users' preferences" (Krishnasamy et al. 2015, p. 445). A common practice of RAs is to apply sponsorship practices and be biased toward sponsored products by manipulating the recommendation order and listing at top place the sponsored products that commonly deviate from the user's preferences (Ransom 2010, Wang et al. 2018).

In the presence of recommendation bias, the users' trust in a biased RA becomes a critical factor determining their use of such RA and their acceptance of the RA recommendations (e.g., Wang and Benbasat 2008, 2016). Especially for a content-filtering-based RA, biased recommendations are not based on the users' explicitly expressed preferences, whereas users normally assume that the RA elicits their preferences to generate recommendations solely according to their expressed preferences (Wang and Benbasat 2008). The provision of biased recommendations may challenge users' trust in the RA and particularly their perceived integrity of the RA. RA integrity is the users' belief that the RA adheres to a set of principles (e.g., honesty and keeping promises) generally accepted by users (McKnight et al. 2002,

Wang and Benbasat 2008). Integrity is a morality-related belief and a key element of users' trust beliefs (Mayer et al. 1995, Ho and Chau 2013). People who behave immorally would be judged as immoral, as people with high morality are presumed to refrain from engaging in immoral behavior (Reeder and Spores 1983). Once a user becomes concerned about the trustee's integrity, she is likely to generate a dispositional attribution of her low integrity belief, and such an attribution is usually stable and difficult to change (Kim et al. 2006, Tomiuk and Pinsonneault 2009). Unlike integrity, other trust beliefs, such as competence, are less stable and could be revised with some effort of the trustee. For instance, individuals can work to develop skills and expertise to improve others' competence-based trust in them (Wang and Benbasat 2016). Researchers suggest that people should do everything possible to minimize others' concerns on integrity issues from the beginning (Kim et al. 2004, 2006). For a biased RA, the RA provider must ensure users' perceived integrity of the RA before considering other aspects, such as the click-through rate of the sponsored recommendations.

A key consideration of a biased RA related to its integrity is whether the RA should *disclose* the sponsorships and biases involved. Such disclosure improves the RA transparency that may lead to users' perceived RA integrity according to knowledge-based trust theory (Wang and Benbasat 2007). Prior studies primarily examine the effects of explanations provided by a neutral RA for its recommendations (i.e., explanations about organic recommendations—henceforth referred to as "organic explanations" in brief) and confirm their positive effects on users' trust in the neutral RA (Wang and Benbasat 2007).[1] Such explanations improve RA transparency by illustrating the inner workings of the RA in generating its organic recommendations. For biased RAs, explanations include both organic explanations and sponsorship disclosure (i.e., explanations for biased or sponsored recommendations). The latter applies exclusively to biased RAs and is the key focus of this research.

Neutrality advocates have long argued for a clear disclosure of any bias or sponsorship involved in recommendation technologies (Crane 2011). The Federal Trade Commission (FTC) released several guidelines governing the use of sponsorship disclosure, including "Dot Com Disclosures" in 2013.[2] The spirit of these guidelines is that if money or any benefit is involved in any way in the recommendation or endorsement of a product or service, the facts must be disclosed (Fletcher 2017). However, Consumer Reports WebWatch affirmed that many technology providers "seem to do as little as possible to explain their relationship with advertisers."[3] MediaRadar claimed that 37% of publishers do not adhere to the FTC rules for labeling paid results; consequently, users cannot easily differentiate between organic and paid content.[4] Many biased RAs fail to clearly disclose the sponsorships involved, such as TripAdvisor (Sullivan 2002) and RAs in pharmaceutical websites (Liang and Mackey 2011) and in health and financial websites (Stanford et al. 2002). The absence of sponsorship disclosure is prevalent in many unbranded websites (Liang and Mackey 2011) and in countries that lack regulations regarding sponsorships in online recommendation and search results.

The proliferation of biased RAs without clear sponsorship disclosure may be explained by the prevailing industry wisdom that the disclosure reveals an ulterior motive and by findings of extant studies suggesting that sponsorship disclosure may undermine source credibility and hamper campaign effectiveness (Commercial Alert 2005, Jansen et al. 2007). However, related research on electronic word-of-mouth (WOM) marketing demonstrates that the disclosure of sponsorships generates either no effect or surprisingly positive consumer responses primarily because the disclosure may be interpreted as a signal of honesty (e.g., Carl 2008, Abendroth and Heyman 2013). Recommending sponsored products without disclosing the sponsorships can be considered a stealth marketing tactic (Kaikati and Kaikati 2004). Controversy surrounding the ethics of stealth marketing has grown over in recent years, and the use of stealth marketing tactics has attracted wide public attention and critiques (Carl 2008, Pehlivan et al. 2015). The effects of sponsorship disclosure in the context of biased RAs remain unclear. Theoretical understanding of these effects for biased RAs is limited.

This research theorizes the effects of sponsorship disclosure on users' perceived integrity of a biased RA from two theoretical perspectives. First, we extend the literature on explanations for neutral RAs and decision support systems to biased RAs and posit the positive effects of disclosure on perceived RA integrity from a knowledge-based trust perspective (Wang and Benbasat 2007). Nevertheless, this perspective may not be the primary mechanism for the effects of sponsorship disclosure in the context of biased RAs, given the potential negative responses of users once they are informed of the use of sponsorships by a biased RA (Carl 2008). Thus, second, and more important, we apply the psychological contract violation (PCV) theory of trust to reveal when the sponsorship disclosure will be optimally effective in increasing users' perceived integrity of a biased RA. Users delegate the task of finding suitable products to a content-filtering-based RA by specifying their preferences to such an RA. This delegation constitutes an implicit psychological contract between an RA and its users such that the RA will recommend products that faithfully match users' indicated preferences (Robinson 1996, Morrison and Robinson 1997, Pavlou and Gefen 2005). The provision of sponsored recommendations violates such a psychological contract, whereas users may expect the RA to fulfil the contract as its obligation. Thus, per the

PCV theory, sponsorship disclosure may not always lead to users' trust in a biased RA. This prediction goes beyond the knowledge-based trust perspective and helps resolve the different interpretations about the consequence of sponsorship disclosure.

The PCV theory of trust has been well established in the context of employee–employer relationships (e.g., Morrison and Robinson 1997) and in the buyer–seller relationships in an e-marketplace (e.g., Pavlou and Gefen 2005). We intend to examine whether sponsorship disclosure mitigates the perceived PCV of a biased RA. High recommendation neutrality is clearly desirable for users (Crane 2011). However, from the perspective of technology providers, sponsorships are necessary for financing the infrastructures of these technologies and enabling and sustaining many free online services that have rapidly become essential to web users (Ransom 2010, Guijarro et al. 2015). Psychological contracts are unilaterally perceived by users, and users may perceive the PCV differently depending on their background knowledge about the bias and sponsorship practices of RAs in general. Some, but not all, users have prior knowledge about the prevalence of sponsorships used by RAs in general (Ma et al. 2013). Users' awareness of bias practice by RAs in general influences their perceived bias of the focal RA (Xiao and Benbasat 2015) and possibly their expected bias and psychological contract with the RA as well. We thus consider such prior knowledge of users as a contingent factor when theorizing the effects of sponsorship disclosure from the PCV perspective of trust.

The primary objective and contribution of this study is to investigate whether and when sponsorship disclosure leads to users' perceived integrity of a biased RA. From the perspective of knowledge-based trust, sponsorship disclosure provided by a biased RA is expected to lead to positive effects by rendering the RA transparent. However, from the PCV perspective, we need to apply a *contingency* approach to reveal when sponsorship disclosure of a biased RA results in users' perceived RA integrity by considering users' prior knowledge about the prevalence of sponsorships used by RAs in general.

The next section reviews the background of this study. Section 3 develops the hypotheses to be tested. Section 4 describes the research method, and Section 5 reports data analyses and results. We conclude with a discussion of contributions, limitations, and avenues for future research.

2. Background: Biased RA

In line with extant behavioral literature on RAs (e.g., Xiao and Benbasat 2015, Wang et al. 2018), we conducted this study in the context of content-filtering-based RAs. For RAs provided by profit-seeking firms, such firms generate revenues from these technologies to achieve returns on their investment. Sponsorships are a major source of such revenues (PricewaterhouseCoopers 2017). The prevailing practice of sponsored RAs is that when a sponsored product falls in the product category of interest to a user, the sponsored product is listed before organic recommendations, regardless of the fit between the sponsored product and the user's preferences (Animesh et al. 2007, Guijarro et al. 2015). Pay-for-placement is one of the most popular models utilized by RAs (Katona and Sarvary 2010). In this model, potential sponsors bid for the right to list sponsored products at the top place of the recommendation results. The top places in the recommendation results are valuable because users are conditioned to consider the rankings of the recommendations as an indication of the degree of relevance to users' preferences (Dou et al. 2010).

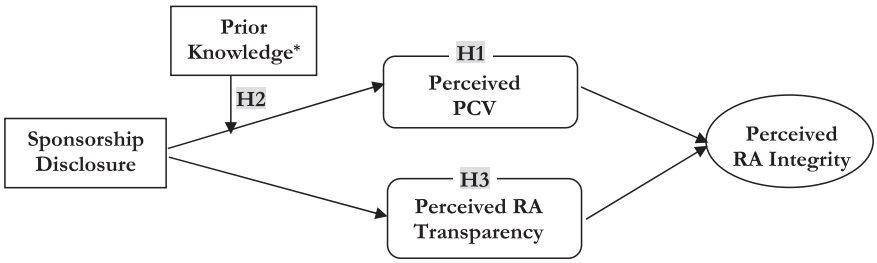
In line with the prevailing practice, we focus on biased RAs that manipulate the recommendation order to favor sponsored products. Sponsored products are not necessarily of low quality but may deviate from users' indicated preferences (Krishnasamy et al. 2015). This type of bias should be salient to users because of such deviation from user preferences. For instance, for a user who prefers Canon cameras (and probably with a few other requirements), an RA biased toward cameras with an unknown brand may recommend the unknown camera as its top recommendation, although such a camera would be among the products that are least likely to be recommended by a neutral RA. In this case, users are highly liable to detect the sponsored products, and such detection may affect the users' evaluations of the RA. Thus, designers must be aware of salient biases and their effects (Krämer et al. 2013, Guijarro et al. 2015).[5] Alternatively, a biased RA may not present sponsored products in the top place and could mix sponsored recommendations with organic ones. This way of presenting sponsored products may get relatively less attention from users but is beyond the scope of this study.

3. Hypothesis Development

Figure 1 shows our research model based on two theoretical perspectives of trust. We first examine the effects of sponsorship disclosure according to PCV theory. A psychological contract is essential in relationship building in an online environment (Rousseau 1995, Robinson 1996, Pavlou and Gefen 2005). PCVs by other entities challenge the foundation of a person's general beliefs regarding others' codes of conduct associated with trusting relationships, thereby reducing the person's integrity belief in others.

In our research context, psychological contracts entail users' beliefs and expectations that RAs should behave in a certain manner based on some implicit assumptions conceived unilaterally by the users (e.g., Robinson 1996). A biased RA normally places sponsored

Figure 1. Research Model



Note. *Users' prior knowledge about the prevalence of sponsorships used by RAs in general.

products as the top recommendations regardless of their relevance to user preferences. However, users generally expect the most relevant products to be listed at the top of the recommendation results given the limited user effort to examine numerous potential recommendations (Dou et al. 2010). Users explicitly indicate their preferences to a content-filtering-based RA as their agent. On the basis of this assumed agency relationship, users may believe that the RA works solely for them and is obligated to provide recommendations that fit their indicated preferences (Wang and Benbasat 2008, Dou et al. 2010, Ma et al. 2013, Xiao and Benbasat 2014). Users may unilaterally establish such a psychological contract with an RA (Rousseau 1995, Robinson 1996, Pavlou and Gefen 2005). PCVs will be perceived when users notice the biased recommendations that deviate from their preferences. The maintenance of psychological contracts is critical to sustaining a trusting relationship (Niehoff and Paul 2001). Perceived PCVs will make users unlikely to believe that a biased RA adheres to a set of basic principles to fulfill its obligations, thereby leading to a low integrity belief in the RA (Niehoff and Paul 2001, Pavlou and Gefen 2005).

However, when a biased RA discloses its use of a sponsorship strategy, the users' implicit assumption that the RA works solely for them is no longer completely valid. Users may have to adjust their expectations slightly regarding what they can receive from the RA and alter their previously unilaterally "conceived" psychological contract with the RA. Users may acknowledge that sponsored recommendations are natural results provided by a biased RA. Users will understand why sponsored products are presented and consider, to a *lesser* degree, the provision of sponsored recommendations as a violation of psychological contract. Thus, we contend that, in general, sponsorship disclosure provided by a biased RA will enhance users' perceived integrity of a biased RA by reducing the perceived PCV of the RA, and we posit the following hypothesis.

Hypothesis 1. *Sponsorship disclosure provided by a biased RA will lead to higher perceived integrity of the RA, and this effect will be mediated by users' perceived PCV of the RA.*

Our argument for the effect of sponsorship disclosure is based on an assumption that the users fully accept the sponsorship practice by the biased RA. However, the extent to which users accept such practice may depend on users' background and prior knowledge about the prevalence of sponsorships used by RAs in general (referred to as "prior knowledge" in brief hereafter). We consider the contingent role of users' prior knowledge and further examine the effect of sponsorship disclosure under two scenarios, one for users with high prior knowledge and the other for users with low prior knowledge.

Users with high prior knowledge typically understand that engaging in sponsorship practices is common for RAs and that sponsorships are necessary to enable and sustain free RA services (Krämer et al. 2013, Ma et al. 2013). These users may not expect the focal RA to exclusively provide organic recommendations or consider the provision of organic recommendations exclusively as essential in their psychological contract with the RA. Their prior knowledge provides an overall knowledge structure to interpret and accept the sponsorship disclosure provided by the RA. The disclosure forewarns users about biases in the RA's recommendations, and users will understand and agree that sponsored recommendations are part of the recommendations provided by the focal biased RA. Thus, users' perceived PCV will be particularly low when they see the disclosure provided by a biased RA. However, when a biased RA does not honestly disclose the sponsorships involved to these users, such users will be skeptical toward the products recommended by the biased RA as they can easily detect biased recommendations given their prior knowledge. These users will have a strong perception of PCV of the biased RA without the disclosure. As such, we contend that the effects of sponsorship disclosure on reducing perceived PCV will be particularly strong for users with high prior knowledge.

For users with low prior knowledge, by contrast, they may be unaware of the common practice of using sponsorships by RAs in general. These users may not

accept this practice when they see the sponsorship disclosure provided by a biased RA. They normally do not expect the focal RA to recommend sponsored products that deviate from their indicated preferences (Wang and Benbasat 2008). Instead, the disclosure may trigger users' consideration of unwanted advertisements of paid results and raise these users' awareness of the "ulterior" motive of the biased RA in recommending sponsored products to them (Carl 2008). These users will consider the provision of sponsored recommendations by the RA as salient PCVs, and their perceived RA integrity remains low, even when the RA discloses the sponsorship involved. That is, sponsorship disclosure will have limited effects on users' perceived RA integrity through the reduced perceptions of PCV. Thus, we postulate the following hypothesis on the effect of sponsorship disclosure on perceived PCV by considering the contingent role of users' prior knowledge.

Hypothesis 2. *The effect of sponsorship disclosure provided by a biased RA on users' perceived PCV of the RA will be contingent upon users' prior knowledge. Such an effect will be stronger for users with high prior knowledge compared with those users with low prior knowledge.*

Prior studies have investigated the effects of explanations provided by computer systems that work solely for users, such as neutral RAs. Wang and Benbasat (2007) investigated the effects of explanations on users' trust in neutral RAs. They affirmed that, in line with the theory of on knowledge-based trust (e.g., Lewicki and Bunker 1995), explanations provided by a neutral RA to illustrate its inner workings can enhance users' understanding of the RA. The RA becomes profoundly transparent with the provision of such explanations, and its behavior becomes predictable. Thus, users' trust in an RA is improved with the increased transparency of the RA when explanations are provided by the RA (Zhou et al. 2018).

In our study, when a biased RA discloses the sponsorships involved in its recommendation generation, users will understand what recommendations are sponsored and how they are produced. These users may perceive the disclosure of sponsorships as a signal of honesty and sincerity, thereby enhancing users' integrity belief in the RA (Forehand and Grier 2003, Abendroth and Heyman 2013, Campbell et al. 2013). In the particular context of RAs, users can iterate their interactions with a biased RA and learn that the biased RA applies the sponsorships as it claims, regardless of their prior knowledge about the use of sponsorships by RAs in general. Users will realize that the biased RA is still predictable in producing its recommendations despite the provision of biased recommendations. The biased RA produces organic recommendations that users could utilize on top of its biased recommendations. Users may consider the RA to have fulfilled its obligation by making it transparent with the disclosure of sponsorships. Consequently, we posit that the following hypothesis on the main effect of sponsorship disclosure.

Hypothesis 3. *The effect of sponsorship disclosure on users' perceived integrity of a biased RA will also be mediated by users' perceived transparency of the RA.*

4. Research Method

We employed a 2 (sponsorship disclosure of a biased RA: with versus without) × 2 (prior knowledge: high versus low) between-subjects, full-factorial experimental design (see Table 1) to test our hypotheses. Neutral RA conditions were included as control groups.

4.1. Experimental RA and Treatments

We designed an experimental RA that was intended to help users choose a digital camera for an experimental decision task (referred to as "camera task" hereafter). The overall design of the experimental RA was consistent with the RAs tested in the literature (e.g., Wang and Benbasat 2009). The experimental RA first elicited users' attribute preferences (e.g., price, brand, resolution) in the form of user–RA dialogue (Komiak and Benbasat 2006, Wang and Benbasat 2009). For most of these attributes, users were provided with three options (levels) to indicate their preferences. They were also asked to indicate the importance levels of these preferences. Users could indicate that they did not have any preference for an attribute. After eliciting these preferences, the neutral RA used an additive compensatory (AC) strategy, a normative decision strategy to generate recommendations based on users' preferences and their importance levels. For biased RAs, the top/first recommendation was a sponsored product. Other than this sponsored recommendation, the products recommended by the biased RA were based on the same strategy applied by the neutral RA. Two sponsored brands were used. One brand was Vivitar, considered an unknown brand of digital cameras in the local community where the experiment was conducted. The other brand was Samsung, a more popular brand than Vivitar.

We validated the experimental RA with the help of several digital camera experts, and their comments were incorporated into the design of the RA. In the experiment, we manipulated recommendation neutrality,

Table 1. Experimental Design

	Prior knowledge	
RA	High	Low
Neutral RA (control group)	Group 1	Group 2
Biased RA with sponsorship disclosure	Group 3	Group 4
Biased RA without sponsorship disclosure	Group 5	Group 5

sponsorship disclosure (for biased RAs only), and users' prior knowledge. A series of pretests was conducted before the main experiment to ensure the effectiveness of these manipulations. Feedback from the pretests was incorporated into the development of the manipulations. These manipulations were further validated in the main experiment. Appendix A provides two screenshots and additional details about the experimental RAs.

4.1.1. Treatment (1): Prior Knowledge. Before the experimental task, we manipulated the participants' prior knowledge by asking (or not asking) them to read an introductory article about pay-for-placement and sponsorship practices used by RAs in practice. The participants assigned to the high prior knowledge condition were requested to read the article supposedly from *Wall Street Journal* (*WSJ*) (see Appendix B). We confirmed in a pretest that *WSJ* was known to most participants and was considered credible. We prepared this article by synthesizing different articles from influential newspapers and magazines.[6] The article explained the wide use of sponsorship practices by extant recommendation technologies and highlighted the pay-for-placement mechanism. After reading the article, the participants in the high prior knowledge condition watched a two-minute recorded video summarizing the article (Dou et al. 2010). Conversely, the participants assigned to the low prior knowledge conditions neither watched this video nor read the *WSJ* article.

4.1.2. Treatment (2): Sponsorship Disclosure. A "Sponsorship Declaration" button was provided by the RA in the "biased RA *with* sponsorship disclosure" conditions (see Figure A.1 in the appendix). Once this button was clicked, the following message appeared: "Please be advised that the first recommended product is a sponsored product. The sponsor paid for this placement in the recommendation list." For the RAs in the "biased RA *without* sponsorship disclosure" conditions, neither the "Sponsorship Declaration" button nor the disclosure information was available to the participants.

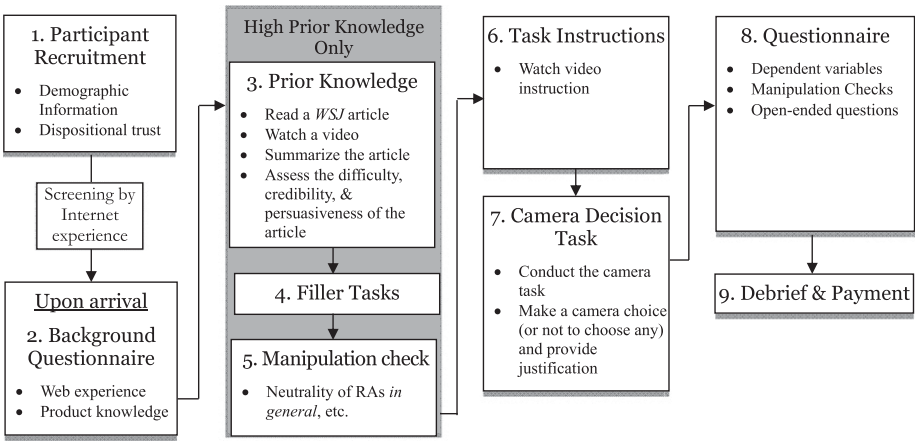
4.2. Participants, Experimental Task, and Procedure

A total of 285 participants were recruited from a public university in Hong Kong to ensure a sufficient statistical power of 0.80 for detecting a small-to-medium effect size (Cohen 1988). Figure 2 illustrates the experimental flow, including the recruitment of the participants. We only recruited participants who have been using the Internet for more than five years to guarantee sufficient web experience for the experimental task. Most participants were in their early 20s. The participants' dispositional trust and some of their demographic information were collected through the online subject recruitment web page. Additional demographic information was collected from the background questionnaire provided upon the participants' arrival for the experiment (Appendix C summarizes the participants' demographic backgrounds).

All experimental sessions were conducted in a laboratory setting, and the participants were randomly assigned to one of the experimental groups. Upon arrival, participants were first surveyed through a background questionnaire about their general web experience and their knowledge of digital cameras (step 2 in Figure 2). No significant differences were found among participants assigned to different experimental groups with respect to these background variables and those collected in the subject recruitment web page.

Steps 3, 4, and 5 applied only to the high prior knowledge conditions. In step 3, after reading the *WSJ* article and watching the video summarizing the key message of the article, the participants were asked to

Figure 2. Experimental Flow



recapitulate the main idea of the article in their own words to determine whether they had read and understood it. The participants also answered three questions concerning the difficulty, credibility, and persuasiveness of the article ("How difficult was the article?"; "How credible was the article?"; and "How persuasive was the article?") using a seven-point Likert scale ranging from 1 ("not at all") to 7 ("very"). In step 4, the participants performed two filler tasks to eliminate short-term memory effects and to preclude experimenter demand effects possibly induced by reading the article (Deval et al. 2013). In step 5, the participants answered a few questions related to their expected neutrality of RAs *in general* to assess their level of agreement with the *WSJ* article, which, in turn, determines the effectiveness of the prior knowledge manipulation.

In step 6, all the participants watched a video that provided instructions on the camera task. The recorded video also showed a tutorial RA, which had the same features as those of the experimental RA later assigned to a particular participant. The video demonstrated the use and navigation of the experimental RA interface. In step 7, all the participants performed the camera task, which involved choosing a digital camera for themselves with the assistance of the experimental RA. This task had no time limit. The task ended either when the participants found a camera and placed it in a shopping cart or when they decided not to choose any of the cameras recommended by the RA. After completing this task, the participants filled out a form to state their product choice and their justification for the choice. In step 8, the participants answered a questionnaire related to our dependent variables (DVs) and manipulation checks. In step 9, the participants were debriefed and received payment for participation in the experiment.

Participation in the experiment was voluntary. Each participant was given HK$50 (approximately US$6.50) for participating in the experiment. The participants were informed before the camera task that 25% of them would receive an additional reward of HK$100 based on their performance as a motivation to take the experiment seriously (Wang and Benbasat 2009). Before the experiment, the participants were advised that they would be asked to justify their product choices and that their performance would be judged on the basis of how convincing their justifications were.

4.3. Measurement

We adapted measurement items in the existing literature for this study. The key variables measured in this study included perceived RA integrity, perceived PCV of the RA, and perceived RA transparency. Appendix D lists all measurement items. The validated measurement scale for integrity was adopted from Vance et al. (2008) and Wang and Benbasat (2007). We adapted the measurement scale for the PCV of online sellers in Pavlou and Gefen (2005) to our context and adapted the measurement items for perceived RA transparency developed by Wang and Benbasat (2016).

To check the manipulation success of prior knowledge, we measured the perceived neutrality of RAs in general after the participants read the *WSJ* article and watched the related video but before they conducted the camera task. We also measured the participants' expected neutrality of the experimental RA after they watched the instruction video for the camera task. The measurement scale for neutrality was adapted from the scales of Tyler (1989) and Colquitt (2001). At the end of the questionnaire provided after the camera task, we asked all the participants whether they had heard of several brand names of digital cameras (including Vivitar) and assessed their level of seriousness in participating in the experiment. A suspicion check was included to determine whether the participants were aware of our actual experimental purpose. We found that none of them was aware of our purposes; thus, data from all the participants were retained for data analyses.

5. Data Analyses and Results

5.1. Manipulation Checks

We assessed the success of the three manipulations (recommendation neutrality, sponsorship disclosure, and prior knowledge). With respect to recommendation neutrality, we first compared the perceived neutrality of the biased RA with the neutral RA via analysis of variance (ANOVA). The ANOVA included three conditions (i.e., neutral RA, biased RA with sponsorship disclosure, and biased RA without sponsorship disclosure). A significant difference was found among the three types of RAs ($M_{\text{Neutral_RA}} = 4.12$, SD = 1.22; $M_{\text{Biased_RA—}\textit{with}\text{—Sponsorship—Disclosure}} = 2.73$, SD = 1.00; $M_{\text{Biased—RA—}\textit{without}\text{—Sponsorship—Disclosure}} = 2.66$, SD = 1.23). Subsequently, we conducted post hoc comparisons of the three RAs with Scheffé's tests. Significant differences ($p < 0.0001$) were observed between (1) biased RA *with* sponsorship disclosure and the neutral RA, and (2) biased RA *without* sponsorship disclosure and the neutral RA. No significant difference was noted between in the two biased RAs (i.e., with versus without the disclosure) ($p = 0.90$).[7] Thus, the neutrality manipulation was successful.[8]

For the *WSJ* article, we checked the participants' summaries of the article and found that most participants adequately encapsulated the article. They considered the article to be not difficult to understand (mean = 3.02, SD = 0.97), credible (mean = 5.58, SD = 0.76), and persuasive (mean = 5.75, SD = 0.77). We examined the participants' level of agreement with the article by calculating their beliefs in the neutrality of

RAs *in general* (mean = 2.60, SD = 0.58; significantly lower than 4, the neutral point on the seven-point scale—note that these data were collected from the participants only after reading the article but before interacting with the experimental RA). This low score demonstrated that the participants assigned to the high prior knowledge condition generally agreed with the article. We also checked the participants' expected neutrality of the experimental RA after receiving the experimental instructions but before interacting with the RA. The article significantly reduced the participants' neutrality expectations of the RA ($M_{High_Prior_Knowledge}$ = 3.51, SD = 1.01; $M_{Low_Prior_Knowledge}$ = 4.03, SD = 1.11; $F = 12.8$, $p < 0.001$). All these results affirmed the success of our prior knowledge manipulation and support our theoretical premise that, for users with different levels of prior knowledge, their psychological contract with the RA differs in connection with the expected neutrality of the RA.

Regarding the sponsorship disclosure manipulation for the biased RAs, the vast majority (91.2%) of the participants clicked the "Sponsorship Declaration" button during their interactions with the experimental RA. We also assessed the self-reported awareness of the sponsorship declaration of the participants assigned to the "biased RA with sponsorship disclosure" condition. These participants were asked to judge the following item in the posttask questionnaire: "This recommendation system informed me about the sponsored product in its recommendations." We found that they were generally aware of the sponsorship declaration (mean = 5.38, SD = 1.22). This result lent support to the sponsorship disclosure manipulation for the biased RAs.

5.2. Measurement Characteristics

We examined the reliability and the validity of the measurement scales. The reliability indicators measured by Cronbach's α (Table 2) were all above the cited minimum value of 0.7 (Nunnally and Bernstein 1994). The discriminant validity of the measurement scales was confirmed using the following two criteria: (1) the square root of the average variance extracted (AVE) of each construct was greater than its correlations with all the other constructs (see Table 2), and (2) the correlation of each item with its intended construct was greater than its correlations with other constructs (see Appendix E) (Barclay et al. 1995). On the basis of these criteria, we concluded that the validity of the measures for our constructs was ensured.

5.3. Hypothesis Testing Results

Table 2 reports the descriptive statistics, including the means and standard deviations of key constructs in different experimental groups. To test the moderating role of prior knowledge (Hypothesis 2) for the effect of sponsorship disclosure on perceived PCV, we conducted an analysis of covariance (ANCOVA) with perceived PCV as the DV and three control variables—namely, disposition to trust, product knowledge, and web experience. Note that this ANCOVA only included biased RA groups because sponsorship disclosure applied only to biased RAs. The ANCOVA results (Table 3) verified that the interaction term was significant. Figure 3 shows the *different* effects of sponsorship disclosure on perceived PCV under these two following conditions: (1) for the participants with *high* prior knowledge (i.e., indicated in the right double arrow in Figure 3) and (2) for those with *low* prior knowledge (i.e., indicated in the left double arrow in Figure 3). The effects of sponsorship disclosure were significant only for the users with high prior knowledge, thereby lending support to Hypothesis 2.

We conducted another ANCOVA with perceived RA transparency as the DV and with the same three control variables. As expected, the results revealed a main effect of the disclosure on perceived RA transparency, and a nonsignificant interaction effect was noted between the disclosure and prior knowledge.

We then tested the mediation roles of perceived PCV (Hypothesis 1) and RA transparency (Hypothesis 3) for

Table 2. Descriptive Statistics: Mean (SD), Correlation, and Reliability

	Mean (SD)									
	High prior knowledge			Low prior knowledge						
Variable	Neutral RA	Biased RA with disclosure	Biased RA without disclosure	Neutral RA	Biased RA with disclosure	Biased RA without disclosure	Correlation 1	Correlation 2	Correlation 3	Cronbach's α
1. *Perceived PCV of RA*	3.43 (1.18)	3.75 (1.13)	4.18 (1.18)	3.16 (1.14)	3.84 (1.14)	3.76 (1.02)	0.87[a]	—	—	0.86
2. *Perceived RA transparency*	4.30 (1.12)	4.12 (1.11)	3.65 (1.22)	4.70 (1.21)	4.27 (0.97)	3.95 (1.18)	−0.49	0.80	—	0.84
3. *Perceived RA Integrity*	4.48 (1.09)	3.90 (1.22)	3.25 (1.23)	4.90 (1.22)	3.97 (1.42)	3.67 (1.20)	−0.76	0.53	0.87	0.85

[a]Diagonal elements are square roots of AVEs, and off-diagonal elements are interconstruct correlations.

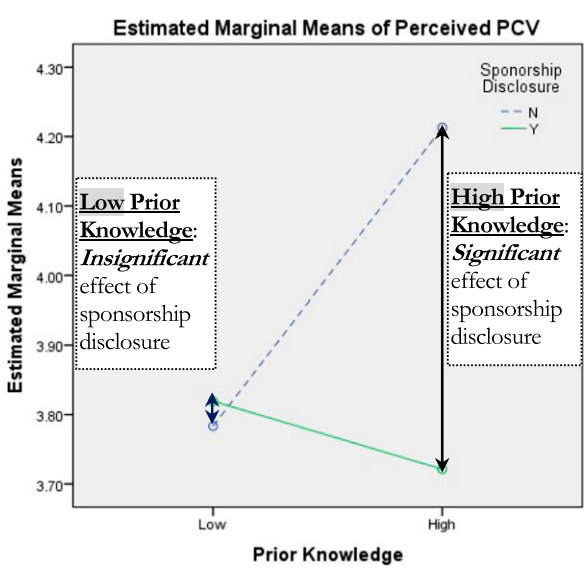
Table 3. ANCOVA Results (DVs: *Perceived PCV* and *Perceived RA Transparency*)

Source	Type III sum of squares	df	Mean square	*F*	Sig.
DV: *Perceived PCV of RA*					
Control: *Disposition to trust*	7.77	1	7.77	6.03	0.02
Control: *Product knowledge*	0.92	1	0.92	0.71	0.40
Control: *Web experience*	0.12	1	0.12	0.10	0.76
Prior Knowledge	1.53	1	1.53	1.19	0.28
Disclosure	2.80	1	2.80	2.18	0.14
Prior Knowledge × Disclosure	5.19	1	5.19	4.23	0.04
Error	282.18	219	1.29		
DV: *Perceived RA Transparency*					
Control: *Disposition to trust*	3.55	1	3.55	2.83	0.09
Control: *Product knowledge*	0.18	1	0.18	0.14	0.70
Control: *Web experience*	1.60	1	1.60	1.28	0.26
Prior Knowledge	4.36	1	4.36	3.48	0.06
Disclosure	4.87	1	4.87	3.89	0.05
Prior Knowledge × Disclosure	2.85	1	2.85	2.27	0.13
Error	274.55	219	1.25		

Note. DV, dependent variable.

the effects of sponsorship disclosure on perceived RA integrity, through a partial least square (PLS) analysis using Smart PLS 2. Following the procedure suggested by Hayes (2013), we tested a mediation model with sponsorship disclosure as the independent variable (IV) (0 = without disclosure, 1 = with disclosure), perceived PCV as the mediator, and integrity as the DV. Consistent with the ANCOVA results, sponsorship disclosure did not exert a significant effect on perceived PCV. As reported previously, the disclosure reduced the perceived PCV only for the participants with high prior knowledge. Thus, we ran the mediation model only for the participants with high prior knowledge. As expected, for these participants, the disclosure reduced perceived PCV ($\beta = -0.18$, $t = 1.99$, $p < 0.05$), which led to higher perceived integrity ($\beta = -0.76$, $t = 17.69$, $p < 0.001$).

Figure 3. (Color online) Interaction Effects of Sponsorship Disclosure and Prior Knowledge on Perceived PCV



Likewise, regarding the role of RA transparency, we tested a mediation model with sponsorship disclosure as the IV, perceived RA transparency as the mediator, and integrity as the DV. We found that the disclosure increased perceived transparency ($\beta = 0.16$, $t = 1.98$, $p < 0.05$), which led to higher perceived integrity ($\beta = 0.56$, $t = 9.15$, $p < 0.001$).

For the participants with high prior knowledge, we also ran a mediation model with both perceived PCV and perceived transparency as the mediators and with the same IV and the DV as the previous models. The effects of the disclosure on perceived PCV and perceived transparency were both significant ($\beta = -0.18$ and 0.22 for these two mediators, respectively), and the effects of the two mediators on integrity remained significant ($\beta = -0.66$ and 0.18 for these two mediators, respectively). The indirect effects of sponsorship disclosure on perceived integrity had a 95% confidence interval (CI) that did not include zero (for perceived PCV, 95% CI = 0.02~0.23; for perceived transparency, 95% CI = 0.08~0.32). After controlling for the mediation processes, the disclosure had no direct effect on perceived integrity ($\beta = 0.07$, $t = 0.99$, $p > 0.05$). These results lent strong support to Hypotheses 1 and 3 on the mediation roles of perceived PCV and RA transparency.

6. Discussion

Drawing on the theoretical perspectives of PCV and the knowledge-based trust, we investigated the effects of sponsorship disclosure of a biased RA on users' perceived RA integrity. Our experimental results affirmed three key findings. First, the effect of sponsorship disclosure on perceived PCV is contingent on whether users possess prior knowledge about the widespread use of sponsorship practice by RAs in general. The disclosure reduced users' perceived PCV of a biased RA only for users who possess high prior knowledge.

A possible alternative explanation could be that instead of the sponsorship disclosure, users' prior knowledge reduced the perceived PCV. This explanation can be ruled out because the main effect of prior knowledge on perceived PCV was *not* significant ($p = 0.28$, according to the ANCOVA results in Table 3). Second, for users with high prior knowledge, the effect of the disclosure on their perceived RA integrity is mediated by the perceived PCV. Third, consistent with prior literature on knowledge-based trust in the context of neutral RAs, the effect of the disclosure on perceived integrity of a biased RA is also mediated by their perceived transparency of the RA, regardless of users' prior knowledge.

6.1. Theoretical Implications

Extant behavioral information systems research on RAs primarily focuses on *neutral* RAs that work solely for users (e.g., Wang and Benbasat 2009). The extant literature provides limited theoretical understanding of the effects of sponsorship disclosure for biased RAs. As an early research endeavor, this study fills gaps in the literature by focusing on *biased* RAs and examining the effects of sponsorship disclosure on gaining users' perceived integrity of biased RAs. In particular, this study has the following two major theoretical implications.

First, this study applies a *contingency* approach to identify a *condition* under which sponsorship disclosure by a biased RA is effective in enhancing users' perceived integrity by reducing their perceived PCV. Controversy surrounds the effects of sponsorship disclosure under different contexts. Effects of disclosure strategies for *biased* RAs require scholarly investigation. This research tackles this controversy in the context of biased RAs from a trust perspective. Biased RAs differ from traditional marketing communications and other persuasive communications for product promotion in online environments, such as WOM marketing. Unlike marketing communications exclusively aiming to promote advertised products, these biased technologies simultaneously provide biased and organic recommendations to users. Although sponsored products may not fit users' preferences well, they normally belong to the same product category of interest to the focal user (e.g., Katona and Sarvary 2010). Biased RAs benefit users by offering organic recommendations while promoting sponsored products (Animesh et al. 2010). Thus, users may have mixed responses to biased RAs, which sequentially offer us with an opportunity to improve users' evaluations through a proper disclosure strategy.

Our research highlights the critical role of users' prior knowledge in theorizing the effect of sponsorship disclosure in the context of biased RAs via the theoretical lens of PCV. Users may possess different levels of prior knowledge on the use of sponsorships by RAs in practice. For users with high prior knowledge, a schema on the bias of RAs *in general* becomes highly accessible during their interaction with a *particular* RA. We confirm that this knowledge base influences users' psychological contract with the focal RA (Levine et al. 2000), as this contract is unilaterally conceived by the user. Only for users with high prior knowledge can they easily accept the sponsorship strategy as disclosed by the focal RA; as a result, the effect of the disclosure on perceived integrity through reduced PCV is effective for these users. We confirmed a moderated mediation relationship for the roles of PCV (mediator) and users' prior knowledge (moderator) for the effect of sponsorship disclosure. Whereas previous research has confirmed that users' awareness of bias practice by RAs in general affects their bias perception of the focal RA (Xiao and Benbasat 2015), to our knowledge, this is the first study to theorize this moderated mediation relationship and the role of prior knowledge for the effect of the sponsorship disclosure.

Second, we extend the knowledge-based trust theory to biased RAs from neutral RAs and knowledge-based systems that work solely for users. Transparency strategies have been considered essential in designing trustworthy technologies (Wang and Benbasat 2007, Granados et al. 2010). This research confirms that even for biased RAs, honestly disclosing the sponsorships involved can still enhance users' perceived integrity.

Trust in RAs can be built through many different processes (Wang and Benbasat 2008). For biased RAs and users with high prior knowledge, the effects of sponsorship disclosure on perceived RA integrity are fully mediated by perceived PCV and perceived transparency. Nevertheless, we revealed some differences in the two theoretical perspectives of trust. For the PCV perspective, sponsorship disclosure enhances perceived integrity through reduced PCV only for the users with high prior knowledge. By contrast, the knowledge-based perspective is always effective in enhancing users' perceived integrity, regardless of users' prior knowledge. To our knowledge, this is the first study that reveals the differences in these theoretical perspectives on trust by considering users' backgrounds in the context of biased RAs.

6.2. Implications for Practice

This study offers two major implications for practitioners. First, practitioners should increase users' awareness of the fact that sponsorships are widely used by RAs in practice to sustain free recommendation services. When users are aware of this practice during their interaction with a biased RA, they will have a relatively more lenient psychological contract with the RA with respect to its biases, and the biased RA can enhance users' perceived integrity through its sponsorship disclosure.

Second, RA providers should follow the guidelines provided by the FTC to faithfully disclose any sponsorships involved to gain users' trust in the RAs.

Examining extant RAs in practice shows that not all biased RAs disclose the sponsorships involved in their recommendations. We asserted that honest disclosure remains the best strategy to improve users' trust in a biased RA. Users may challenge the integrity of a biased RA in the absence of clear disclosure of sponsorships, especially for the biases that can be easily detected by users. Even when users do not notice a biased recommendation during the earlier stage of their interaction with the RA, they may do so after repeated usage of the RA. Third-party monitoring websites also check the neutrality of RAs and report the bias to the public if RAs do not disclose the biases (e.g., ProPublica.org). Prior research has demonstrated that once the integrity is challenged, regaining users' perceived integrity of the RA will be extremely difficult (Kim et al. 2006, Tomiuk and Pinsonneault 2009).

6.3. Limitations and Future Research

This study bears two major limitations and offers several opportunities for future research. First, we addressed the controversy on the effects of sponsorship disclosure only from the perspective of users' trust in a biased RA. Sponsorship disclosure may also affect users' other responses, such as their click-throughs of the sponsored recommendations and their attitude toward the sponsored products. Future research is required to examine different outcomes of sponsorship disclosure related to these outcome variables.

Second, the effects of the sponsorship disclosure were examined empirically in our specific RA context and with specific manipulations as operationalized in our experiment. The bias manipulation was made salient in that sponsored products were not mixed with organic recommendations and possibly conflicted with users' explicitly indicated brand preferences. Biases could be made subtler. For instance, the RA can be biased toward brands that still fit users' preferences or toward factors unrelated to users' indicated preferences, or the RA may mix the sponsored recommendations with organic ones. Users may be unable to identify such biases during their first interaction with the RA. Readers should be cautious when generalizing our results to the contexts of subtle biases. Future research should investigate the effects of disclosure of these subtle biases, although these biases may not necessarily involve sponsorships.

With respect to the prior knowledge manipulation, we did not rely on the participants' actual prior knowledge because of the relatively homogeneous background of the participants in a laboratory setting. Instead, we carefully designed the manipulation to introduce different levels of "prior" knowledge before the experimental task. Future research may be conducted in a field setting in which a wider range of prior knowledge levels exists.

This research was also conducted in the specific context of content-filtering-based RAs with faithful disclosure of sponsorships involved. Other types of RAs exist, such as collaborative-filtering-based RAs that do not explicitly elicit users' preferences. The type and level of biases involved in these RAs may be different from the RAs studied in this study and require future research. Moreover, we manipulated sponsorship disclosure for biased RAs only. Indeed, a neutral RA could explain to the users that it does not involve sponsorships, and a biased RA may deceive the user by explaining so. The effect of such disclaims might be different from those of the faithful disclosure of a biased RA, and the latter case involves deceptions in addition to recommendation bias. Future research is needed in these areas.

Fit scores were provided for organic recommendations only, but not the sponsored product, in our experiment. Nevertheless, a biased RA could calculate the fit score for the sponsored product. The effects of sponsorship disclosure may differ depending on the relative value of the fit score of the sponsored product compared with the top organic recommendations. We call for future research to investigate this issue.

7. Conclusion

Sponsorship disclosure is a central issue for biased RAs. From a PCV perspective, we apply a contingency approach to study the effect of the disclosure on users' perceived integrity of biased RAs. Sponsorship disclosure can enhance perceived integrity via reduced perceived PCV only for users with high prior knowledge but not for those with low prior knowledge. From a knowledge-based trust perspective, we support the general principle of "honesty is the best policy" and confirm the positive effect of the disclosure through the increased transparency of the RA. This study not only advances the theoretical understanding of the effects of sponsorship disclosure on trust in RAs but also offers guidelines for practitioners to offer biased, albeit trustworthy, RAs. With trust, biased RAs can provide utilities to users such that they can utilize the recommendations (especially the organic ones), whereas the RA providers can generate profit from applying sponsorship practices, thereby affording a win–win situation for both the users and the providers of biased RAs.

Acknowledgments

The authors thank senior editor Paul A. Pavlou, associate editor Animesh Animesh, and the three anonymous reviewers

for excellent comments and constructive feedback that have significantly improved the quality of the paper.

Appendix A. The Experimental RA

The neutral RA in the laboratory experiment employed an AC strategy. The AC-based RA has been validated as an effective recommendation technology (Song et al. 2007, Wang and Benbasat 2009). It evaluates each product alternative based on all its relevant attributes, and each attribute is assigned an importance weight by the user. A fit score was determined for each product alternative, and users are provided with a list of 10 recommended alternatives in descending order of the fit score.

For biased RAs, we also manipulated the attribute levels of the two "sponsored" products to ensure that these two cameras should be among those least likely to be recommended according to users' preferences (i.e., with the lowest fit). For a neutral RA, these two sponsored products would never be listed as top recommendations. In the absence of this manipulation of product attributes, the two sponsored products might be actually recommended as a top organic recommendation to users because the users might indicate many other attribute preferences apart from brand. In this case, no difference would exist in the recommendation results between a biased and a neutral RA. Thus, this manipulation of product attributes further ensured the existence of recommendation bias for the biased RA.

Appendix B. Fictitious Article from *The Wall Street Journal*

Marketers Should Pay to Play—Search Engine Poses Biggest Threat Yet to $100-Billion Classified Ad Industry

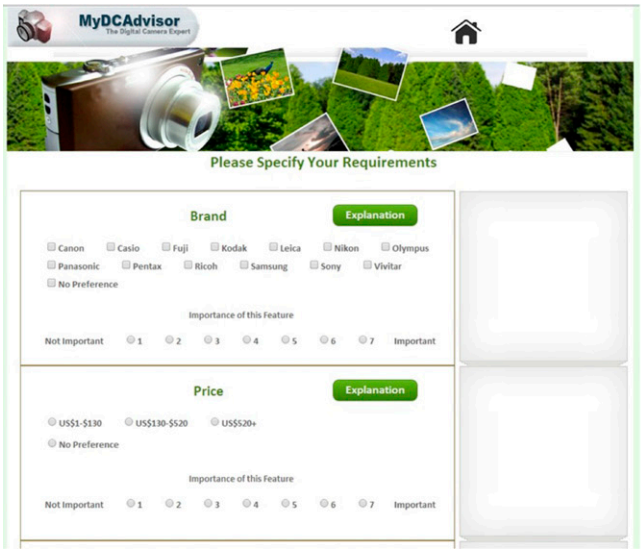
Today, more and more consumers use search engines (e.g., Google, Yahoo, Baidu) to find products and companies of their interest. However, many search engines (e.g., Baidu) no longer provide "neutral" search results. To reach more customers, companies can pay for placement (i.e., to be listed at the top of the research results) to achieve or maintain high rankings in search engines. This is called pay-for-placement.

Pay-for-placement is an internet advertising model. Under this model, advertisers bid for the right to present an advertisement, and higher ranking bids appear at the top of the page. When pay-for-placement was first introduced, 70% of internet users were unaware that search results could be skewed as a result of such agreements. However, pay-for-placement is now seen as the most efficient and effective way to make profits for search engines. Yahoo pioneered the "pay-for-placement" model for online advertising in 2003. Net advertising enabled Yahoo to double its earnings in the first quarter of 2004.

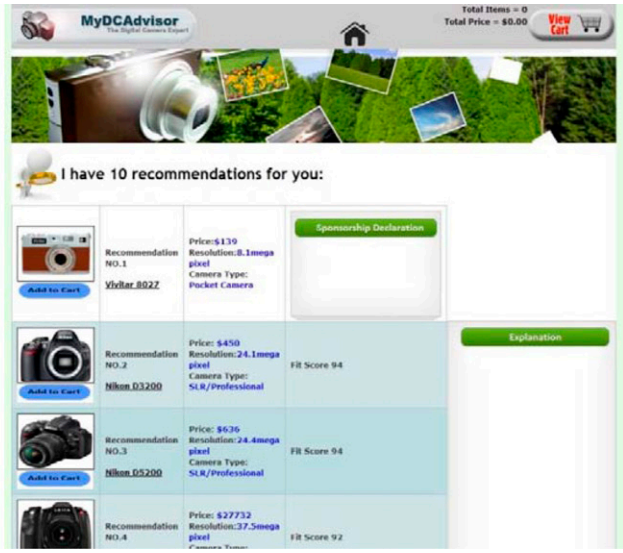
"If companies need a traffic jolt, they can pay for placement in search engines," commented Catherine Seda.

Catherine Seda is president of Seda Communication (SedaCommunication.com), an internet marketing and training company, and a popular conference speaker.

Figure A.1. (Color online) RA Screenshots



Questions from the RA



Recommendation Results Page
(Experimental Condition: *Biased* RA with sponsorship disclosure)

Appendix C. Demographic Background of Experimental Participants

Variable/item		No. of participants (%)	Mean
Age		—	21.8
Gender	Male	120 (42.1%)	—
	Female	165 (57.9%)	—
Web Experience			
How often did you shop online (auctions, group buy, Taobao, etc.) in the past 12 months?	Never	0	—
	1–6 times	207 (72.6%)	—
	7–12 times	44 (15.4%)	—
	12–24 times	22 (7.7%)	—
	More than 24 times	12 (4.2%)	—
How many years have you used the internet?	Less than 1 year	0	—
	1–2 years	0	—
	3–4 years	0	—
	5–7 years	35 (12.3%)	—
	More than 7 years	250 (87.7%)	—
On average, how much time per day do you spend on each of the following web activities: Accessing information on the web about products and services you may buy?	Never	0	—
	0–30 minutes	164 (57.5%)	—
	30–60 minutes	72 (25.3%)	—
	1–2 hours	36 (12.6%)	—
	2–3 hours	13 (4.6%)	—
	More than 3 hours	0	—
On average, how much time per day do you spend on each of the following web activities: Shopping (i.e., actually purchasing something) on the web?	Never	0	—
	0–30 minutes	238 (83.5%)	—
	30–60 minutes	33 (11.6%)	—
	1–2 hours	12 (4.2%)	—
	2—3 hours	2 (0.7%)	—
	More than 3 hours	0	—

Appendix D. Measurement Scales

The measurement items of the following three variables all appeared in the questionnaire after the participants finished the camera decision task. Participants rated the items on a seven-point Likert scale, ranging from 1 ("strongly disagree") to 7 ("strongly agree"), except where indicated.

(1) Perceived RA integrity (Wang and Benbasat 2007, Vance et al. 2008)

- *INTGR1*: This recommendations to me provided by the recommendation system were truthful.
- *INTGR2*: I would characterize this recommendation system's dealing with me as honest.
- *INTGR3*: This recommendation system appeared to be unbiased.

(2) Perceived PCV of RA (Pavlou and Gefen 2005)

- *PCV1*: The recommendation system failed to meet its obligations to you during your interaction.
- *PCV2*: The recommendation system did a good job of meeting its obligations to you during your interaction. (Reversed)
- *PCV3*: The recommendation system fulfilled most important obligations to you during your interaction. (Reversed)

(3) Perceived RA Transparency (Wang and Benbasat 2016)

- *TRNSP1*: This recommendation system made its reasoning process clear to me.
- *TRNSP2*: I could easily understand this recommendation system's reasoning process.
- *TRNSP3*: It was easy for me to understand the inner workings of this recommendation system.
- *TRNSP4*: This recommendation system's logic in providing advice was clear to me.
- *TRNSP5*: This recommendation system hid important information that might reflect badly on the virtual advisor. (Reversed)

The following three constructs were measured for the manipulation check purposes.

(1) Neutrality of RAs in general (Tyler 1989, Colquitt 2001): The following items appeared only in the questionnaire given to the participants assigned to the high "prior knowledge" conditions. These participants were asked the following items before they performed the camera decision task.

- *NTRLT_Gnrl1*: Many online product recommendation systems favor their sponsors' products over others in their recommendations. (Reversed)
- *NTRLT_Gnrl2*: Many online product recommendation systems are influenced by their sponsors' products when providing product recommendations. (Reversed)
- *NTRLT_Gnrl3*: Online product recommendation systems are free of bias when providing product recommendations.
- *NTRLT_Gnrl4*: Many online product recommendation systems provide misleading product recommendations. (Reversed)

(2) Neutrality expectation of the experimental RA (Tyler 1989, McKinney et al. 2002): All participants were asked these items *after* they watched the video instructions for the camera decision task and the experimental RA but *before* they performed the camera decision task.

On the basis of the information and video shown to you about the recommendation system, do you expect the camera recommendation system

- *EXPCT1*: to favor certain camera vendor(s) that you may not prefer? (Reversed)
- *EXPCT2*: to be influenced by other factor(s), besides your preferences, when providing camera recommendations? (Reversed)
- *EXPCT3*: to be free of bias, except for your preferences, when providing camera recommendations?
- *EXPCT4*: to provide misleading camera recommendations? (Reversed)

(3) Perceived recommendation neutrality of the experimental RA (Tyler 1989, Colquitt 2001)

- *NEUTR1*: This recommendation system did not favor any camera vendor except for my preferences.
- *NEUTR2*: This recommendation system was influenced by other factor(s), besides my preferences, when providing camera recommendations. (Reversed)
- *NEUTR3*: This recommendation system was free from bias, except for my preferences, when providing camera recommendations.
- *NEUTR4*: This recommendation system provided misleading camera recommendations. (Reversed) (This variable was dropped.)

Appendix E. Measurement Validation

The correlations among measurements items and those among the items and the constructs are shown in Table E.1.

Table E.1. Correlations Among Items and Constructs

	Integrity	PCV	Transparency	1	2	3	4	5	6	7	8	9	10	11
1. *INTGR1*	0.88	−0.62	0.54	1										
2. *INTGR2*	0.91	−0.63	0.51	0.78	1									
3. *INTGR3*	0.81	−0.54	0.33	0.73	0.77	1								
4. *PCV1*	−0.30	0.85	−0.35	−0.55	−0.58	−0.43	1							
5. *PCV2*	−0.73	0.89	−0.49	−0.59	−0.59	−0.29	0.62	1						
6. *PCV3*	−0.67	0.87	−0.44	−0.55	−0.52	−0.47	0.69	0.69	1					
7. *Transparency1*	0.44	−0.40	0.88	0.46	0.41	0.27	−0.31	−0.39	−0.34	1				
8. *Transparency2*	0.43	−0.39	0.89	0.45	0.41	0.26	−0.26	−0.40	−0.35	0.80	1			
9. *Transparency3*	0.32	−0.30	0.83	0.29	0.31	0.22	−0.21	−0.32	−0.25	0.65	0.72	1		
10. *Transparency4*	0.51	−0.50	0.83	0.52	0.51	0.31	−0.38	−0.45	−0.48	0.67	0.68	0.60	1	
11. *Transparency5*	0.34	−0.30	0.72	0.38	0.32	0.19	−0.18	−0.32	−0.27	0.61	0.61	0.71	0.71	1

Note. Correlations are all significant at the 0.01 level (two-tailed).

Endnotes

[1] Likewise, the literature on decision support systems and knowledge-based systems in general has attested to the positive effects of the explanations provided by these systems that work solely for the users (Gregor and Benbasat 1999).

[2] See https://www.ftc.gov/sites/default/files/attachments/press-releases/ftc-staff-issues-guidelines-internet-advertising/0005dotcomstaffreport.pdf, accessed August 31, 2017.

[3] See http://www.pewinternet.org/2005/01/23/part-5-is-users-trust-in-search-engines-blind/, accessed August 31, 2017.

[4] See https://resources.mediaradar.com/newsroom/native-advertising-continues-to-grow-but-compliance-issues-remain, accessed August 31, 2017.

[5] Some RA biases may not be so salient to users when the RA considers factors aside from those related to users' indicated preferences to determine its recommendation order, or when the RA mix the sponsored recommendations with organic ones. Nevertheless, users may detect these biases sooner or later through repeated use or learning from others' experience. Online word of mouth travels fast in the internet age. Aside from traditional consumer reports, monitoring websites (e.g., ProPublica.org) have reported many subtle, undisclosed biases (e.g., Staples offered the same products at higher prices to people living in a certain ZIP code; see https://www.propublica.org/article/breaking-the-black-box-when-algorithms-decide-what-you-pay, accessed January 9, 2018).

[6] Two major reference sources of the article were (1) "Yahoo retools ad technology; ranking system ends pay-for-placement ads in search results" (*Washington Post*; February 6, 2007) and (2) "Marketers should pay to play—Why search engine optimization is not enough" (*PR Newswire*; October 25, 2004).

[7] We also compared the differences in the participants' evaluations of the RA biased to Samsung versus the RA biased to Vivitar. We conducted a multiple analysis of variance (MANOVA) for these two conditions to check whether the different brands (i.e., Samsung and Vivitar) exerted an overall effect on users' perceived neutrality and RA integrity. The MANOVA results revealed no significant difference between the RAs biased to the two brands ($F = 2.56$; $p > 0.05$). Thus, we combined these two biased RA conditions into one (i.e., the biased RA condition) in the remaining analyses.

[8] We also checked whether the participants in the biased RA conditions actually identified the biases based on our intended manipulation of bias toward the two sponsored camera brands. At the end of the posttask questionnaire, we asked an open-ended question on whether the participants had encountered any anomaly in the received recommendations. We found that the majority of the participants (73%) in the biased conditions mentioned the Samsung or Vivitar camera in their answers to the open-ended question as anomalies because of the deviation of these two products from their brand preferences. This result further supported the success of our intended neutrality manipulation.

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